

Lab – Digital Forensics Analysis Using Autopsy

Overview

In a previous lab, we installed Autopsy onto a standalone virtual Windows 10 Pro machine. In this lab, we will see how we create a case in Autopsy, import a forensic image, and conduct a forensic analysis of a suspect's computer.

Lab Requirements

One Windows 10 Pro machine (Virtual or otherwise)

One install of Autopsy

One copy of the forensic image file, [4Dell Latitude CPi.E01](#)

Lab Scenario

It is suspected that this computer was used for hacking purposes, although it cannot be tied to a hacking suspect, Greg Schardt. Schardt also goes by the online nickname of “Mr. Evil,” and some of his associates have said that he would park his vehicle within range of Wireless Access Points where he would then intercept internet traffic, attempting to get credit card numbers, usernames & passwords. Find any hacking software, evidence of their use, and any data that might have been generated. Attempt to tie the computer to the suspect, Greg Schardt.

Begin the Lab!

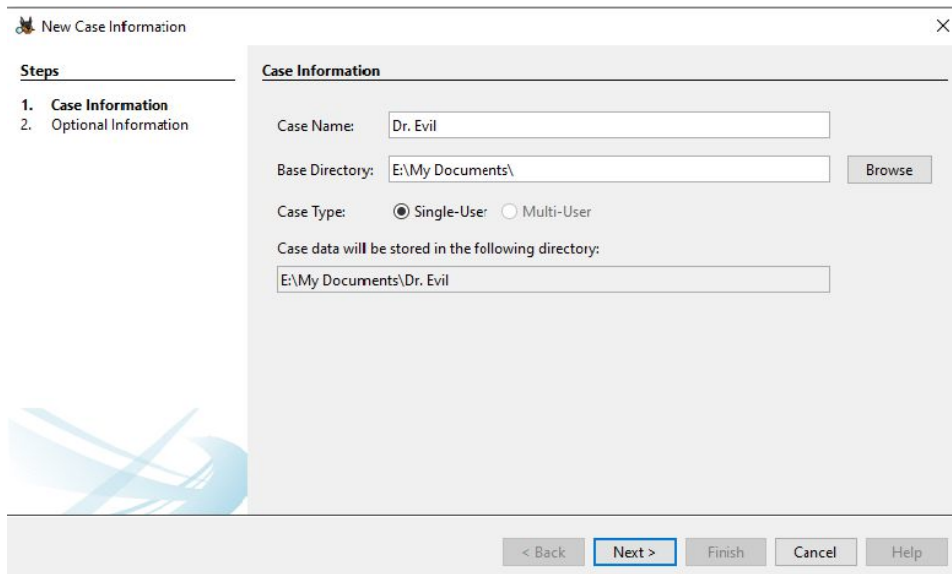
Launch Autopsy either for your virtual installation of Windows 10 or your personal device.

Once Autopsy loads, click the option to create a new case.

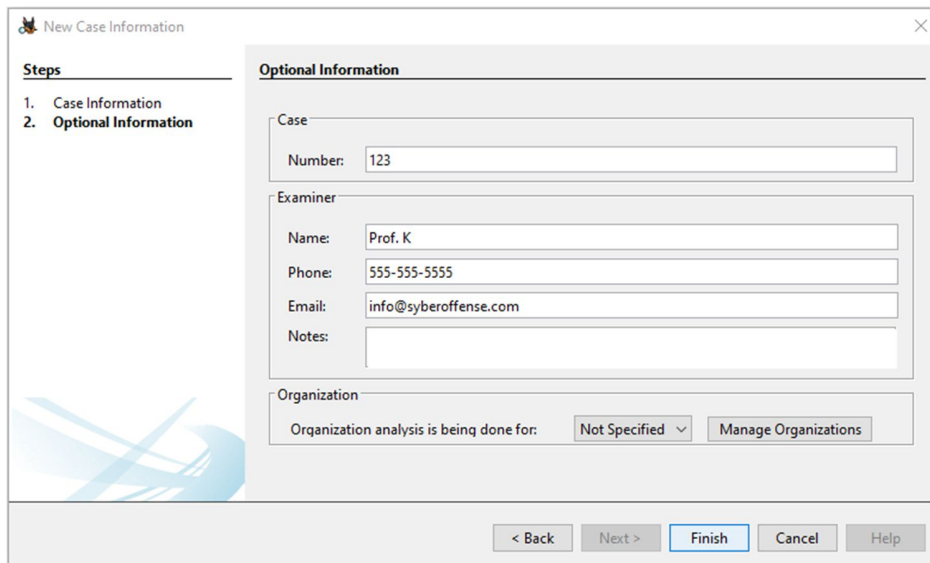


Provide the Case Name and the directory to store the case file. Click on Next.

You are free to name your case as you see fit.

The image shows the 'New Case Information' dialog box. On the left, there is a 'Steps' section with two items: '1. Case Information' and '2. Optional Information'. The 'Case Information' section is currently active. The main area of the dialog is titled 'Case Information' and contains several input fields and buttons. The 'Case Name' field is filled with 'Dr. Evil'. The 'Base Directory' field is filled with 'E:\My Documents\' and has a 'Browse' button next to it. The 'Case Type' section has two radio buttons: 'Single-User' (which is selected) and 'Multi-User'. Below this, there is a text box that says 'Case data will be stored in the following directory:' followed by a text field containing 'E:\My Documents\Dr. Evil'. At the bottom of the dialog, there are five buttons: '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'. The 'Next >' button is highlighted with a blue border.

Add Case Number and Examiner's details, then click on Finish.



New Case Information

Steps

1. Case Information
2. **Optional Information**

Optional Information

Case

Number: 123

Examiner

Name: Prof. K

Phone: 555-555-5555

Email: info@syberoffense.com

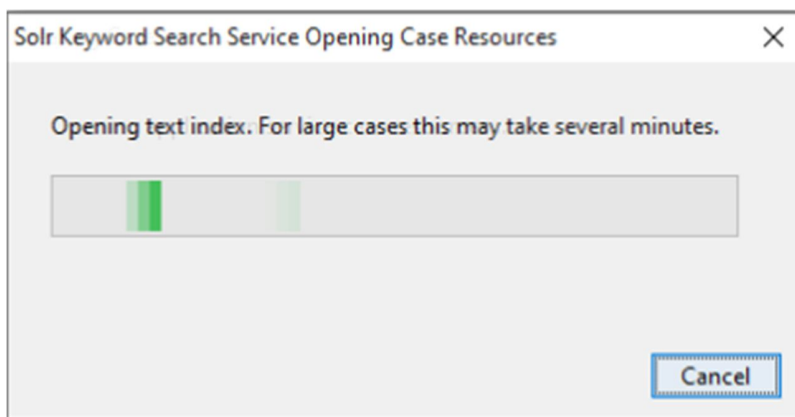
Notes:

Organization

Organization analysis is being done for: Not Specified Manage Organizations

< Back Next > Finish Cancel Help

Autopsy generates a case file.

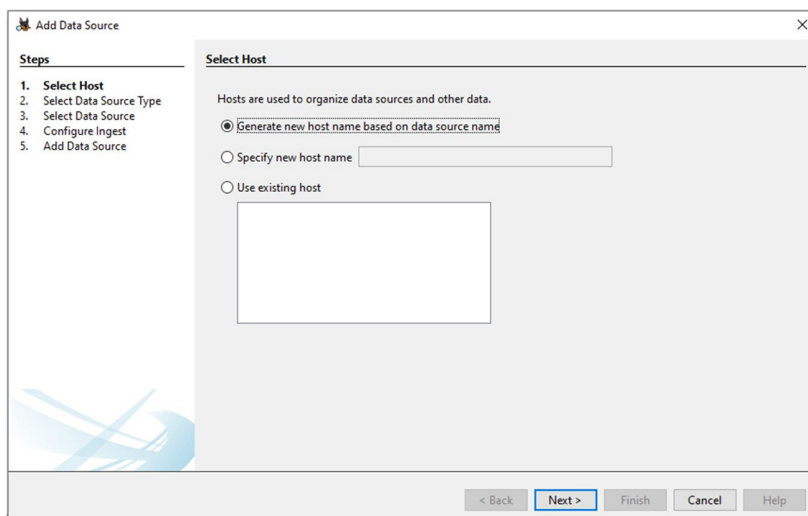


Solr Keyword Search Service Opening Case Resources

Opening text index. For large cases this may take several minutes.

Cancel

On the Add Data Source page, accept the defaults and click next.



Add Data Source

Steps

1. **Select Host**
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source

Select Host

Hosts are used to organize data sources and other data.

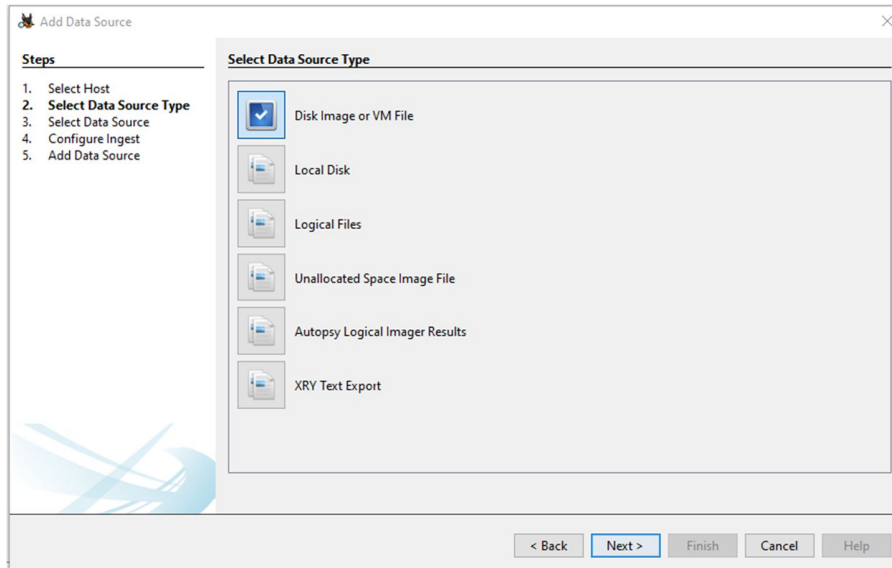
☒ Generate new host name based on data source name

☐ Specify new host name

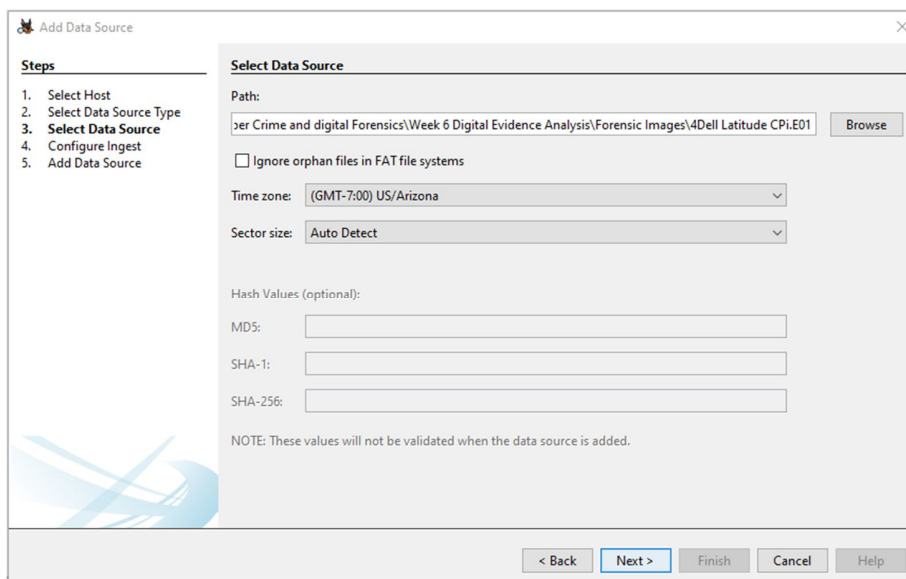
☐ Use existing host

< Back Next > Finish Cancel Help

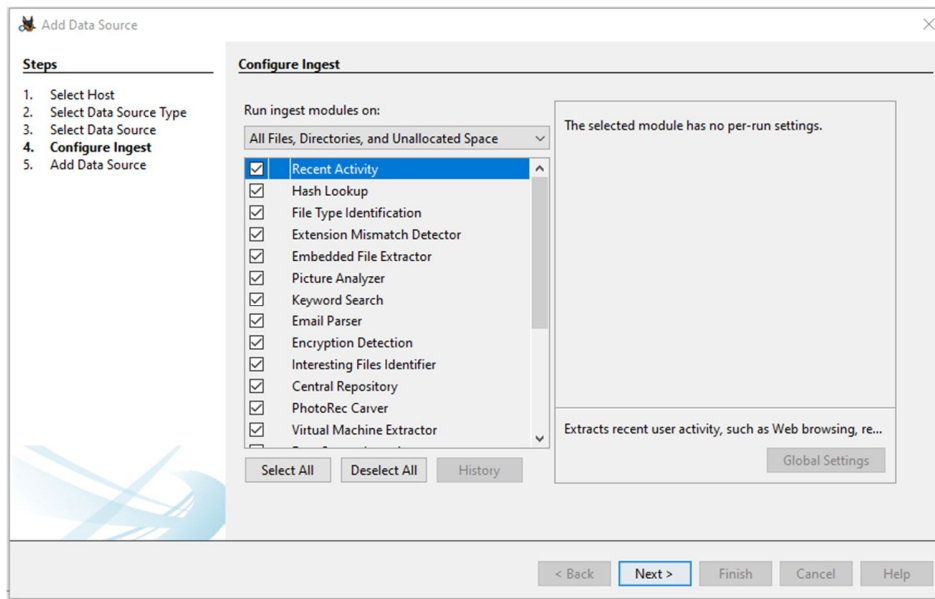
Choose the required data source type, in this case, Disk Image, and click on Next.



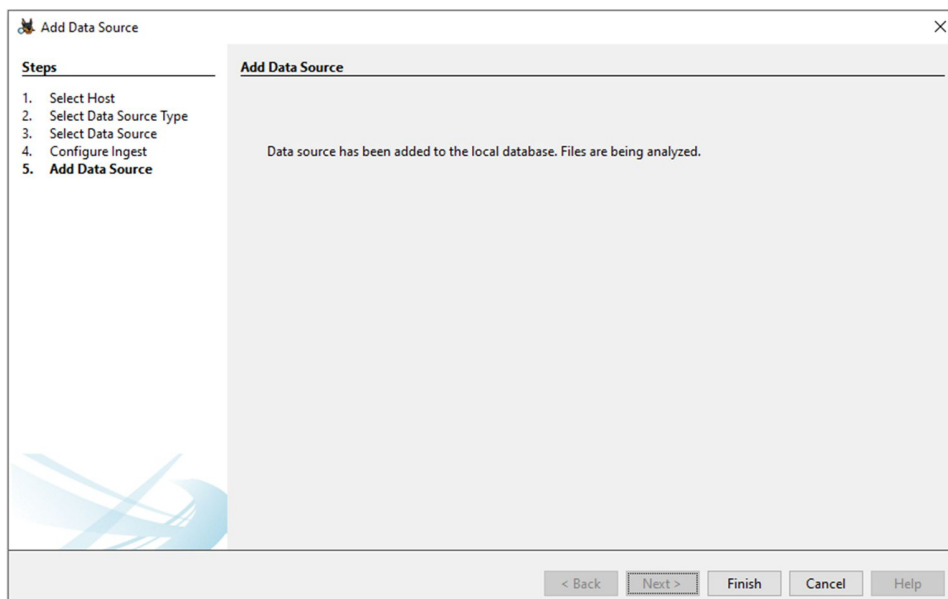
Give path of the data source and click on *Next*.



Accept the default modules that are checked and click Next.



After the data source has been added, click on Finish.



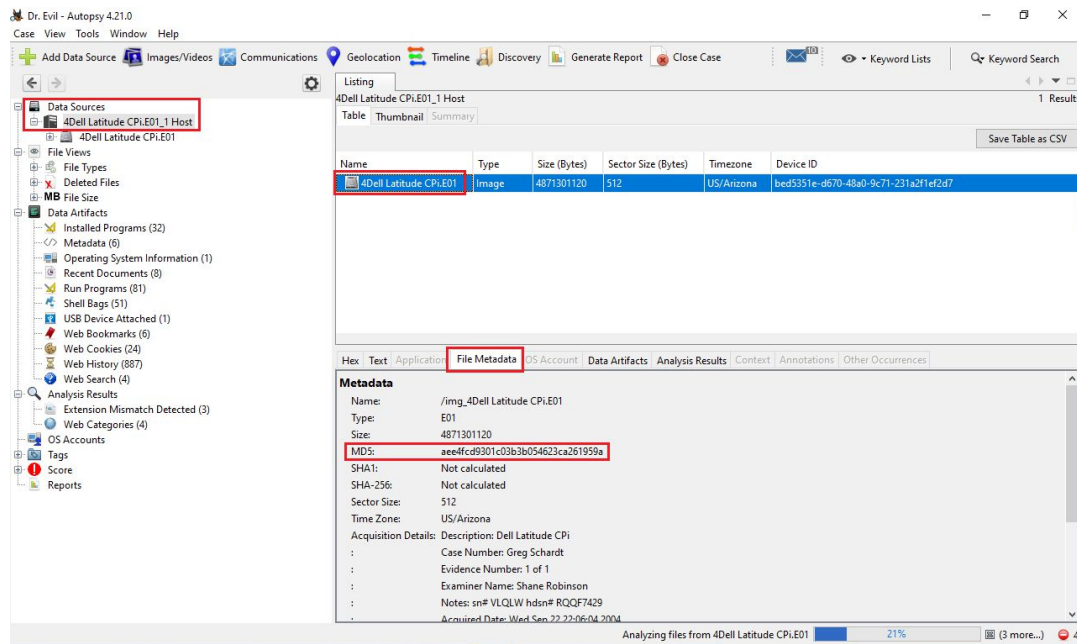
Once all the modules have been ingested, you can begin investigating. It is recommended waiting until the analysis and integrity check are complete.

Begin your analysis by answering the following questions.

Q1. What is the image hash?

Answer: AEE4FCD9301C03B3B054623CA261959A.

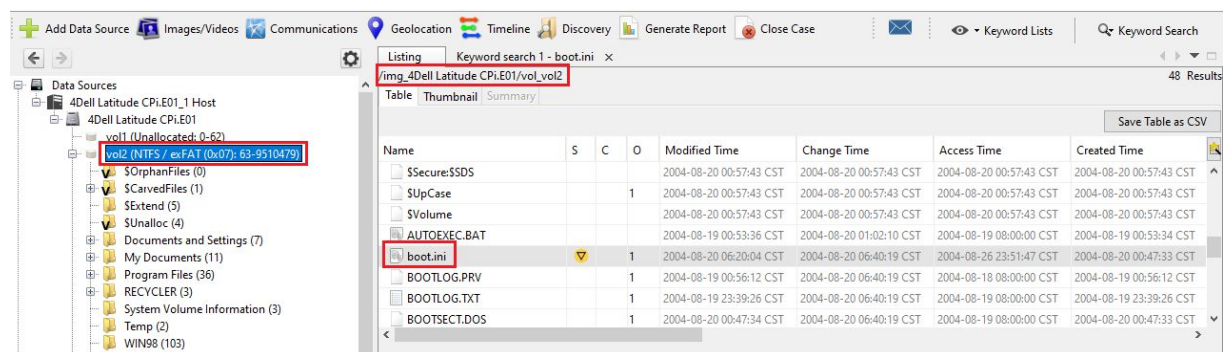
To check the image hash, click on the image and go to the File Metadata tab. (We check the image hash to verify that it is the same as the hash created during the time when the image was created.)

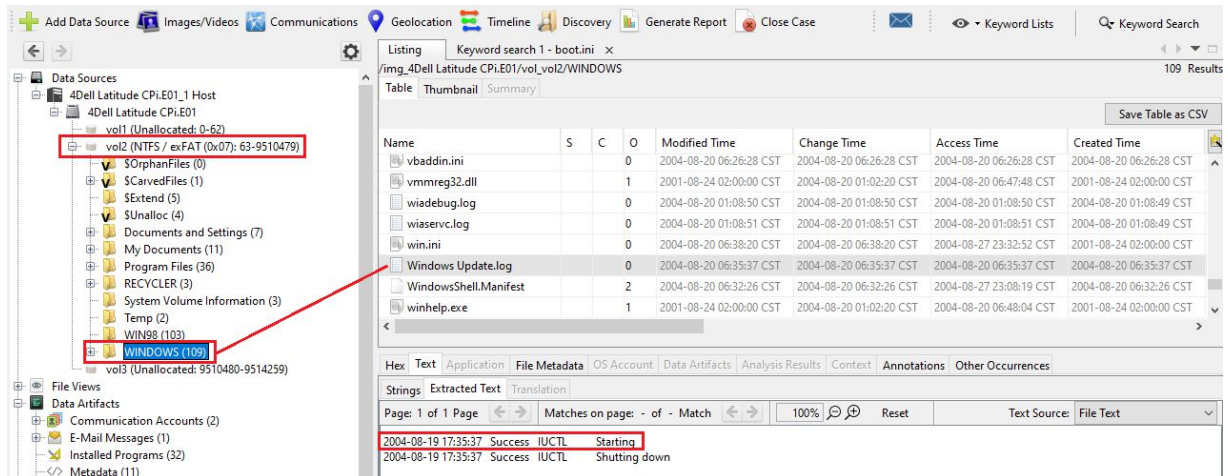


Q2: What operating system was used on the computer?

Answer:

Look in the C:\boot.ini file.

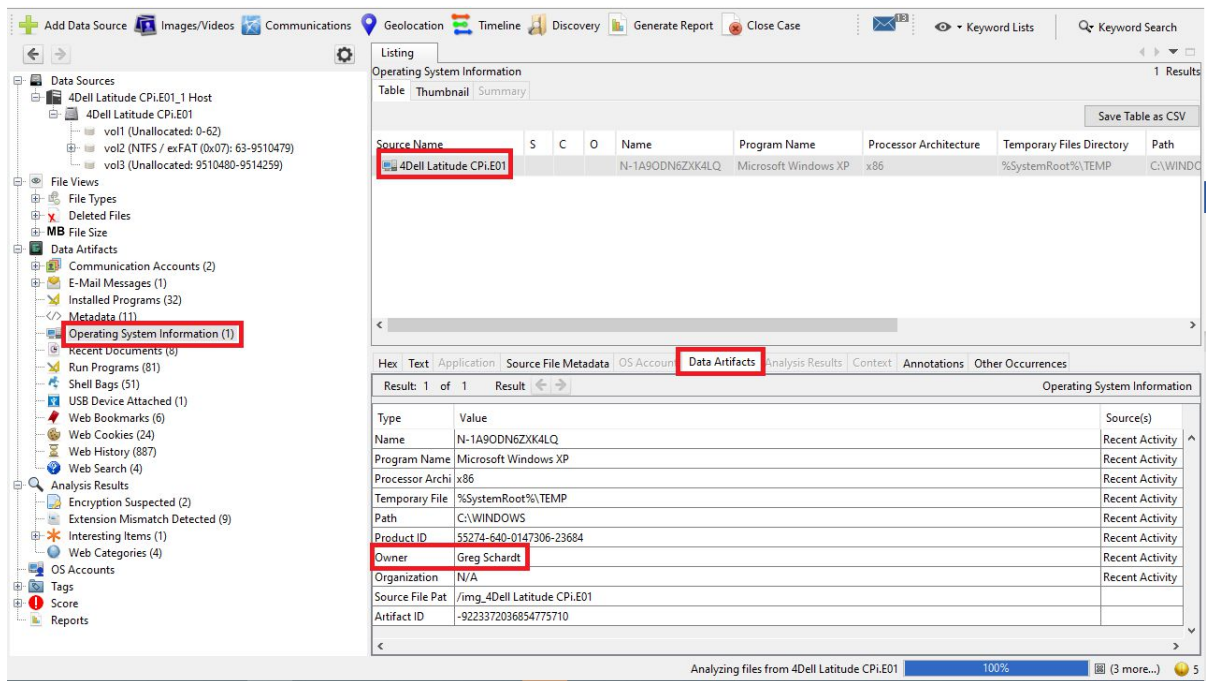




Q4. Who is the registered owner?

Answer. Greg Schardt

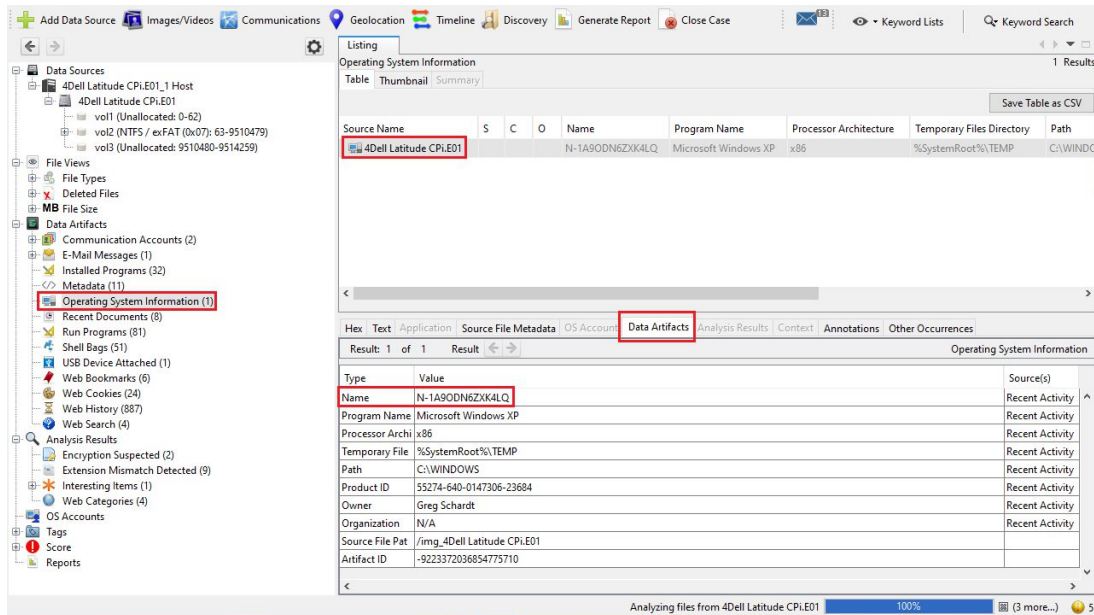
Answer. (Click on *Operating System Information*)



Q5. What is the computer account name?

Answer. The account name is N-1A9ODN6ZXK4LQ

(Click on *Operating System Information*)

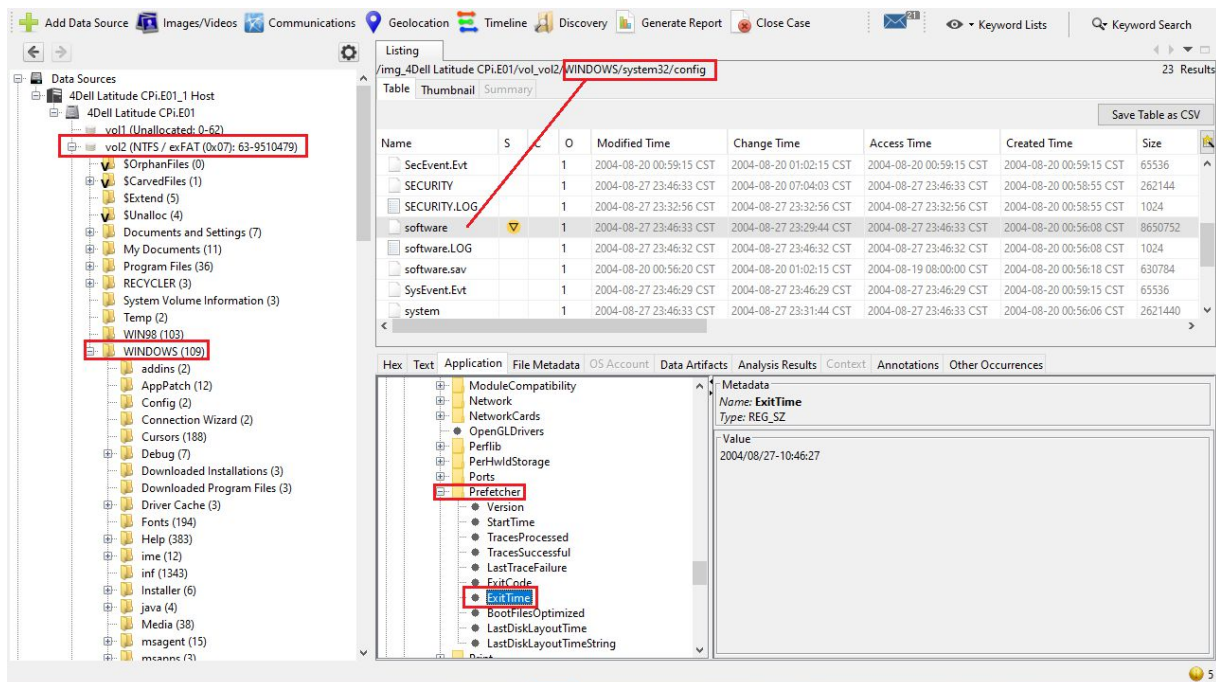


Q6. When was the last recorded computer shutdown date/time?

Answer: The last recorded shutdown time of the computer is 2004/08/27-10:46:27

To find this, go to:

C:\WINDOWS\system32\config\software\Microsoft\WindowNT\CurrentVersion\Prefetcher\ExitTime



Q7. How many accounts are recorded (total number)?

Answer: 5 accounts, Mr, Evil, Administrator, Guest, Support388945a0, and HelpAssistant

In the left side panel, we go to Analysis Results > OS Account

The screenshot shows a forensic analysis tool with a left sidebar and a main panel. In the sidebar, 'Analysis Results' and 'OS Accounts' are highlighted with red boxes. A red arrow points from 'OS Accounts' to the main panel. The main panel displays a table of OS accounts.

Name	S	C	O	Login Name	Host	Scope	Realm Name	Creation Time
S-1-5-18				SYSTEM	4Dell Latit...	Local	NT AUTHORITY	
S-1-5-21-2000478354-688789844-1708537768-1002			0	SUPPORT_388945a0	4Dell Latit...	Local	NT AUTHORITY	2004-08-20 06:35:19 CST
S-1-5-20				NETWORK SERVICE	4Dell Latit...	Local	NT AUTHORITY	
S-1-5-21-2000478354-688789844-1708537768-1003			0	Mr. Evil	4Dell Latit...	Local	NT AUTHORITY	2004-08-20 07:03:54 CST
S-1-5-19				LOCAL SERVICE	4Dell Latit...	Local	NT AUTHORITY	
S-1-5-21-2000478354-688789844-1708537768-1000			0	HelpAssistant	4Dell Latit...	Local	NT AUTHORITY	2004-08-20 06:28:24 CST
S-1-5-21-2000478354-688789844-1708537768-501			0	Guest	4Dell Latit...	Local	NT AUTHORITY	2004-08-20 00:59:24 CST
S-1-5-21-2000478354-688789844-1708537768-500			0	Administrator	4Dell Latit...	Local	NT AUTHORITY	2004-08-20 00:59:24 CST

Q8 .Who was the last user to logon to the computer?

Answer:: Mr. Evil

The system will obtain the last user who logged on from the key 'DefaultUserName'. This information can be uncovered from the following path

Click on Data Sources select 4Dell Latitude --> vol2 -->

WINDOWS/SYSTEM32/CONFIG/SOFTWARE/MICROSOFT/WINDOWS
NT/CURRENT

VERSION/WINLOGON/DEFAULT USER NAME

Q9 . List the network cards used by this computer.

Answer:

- Compaq WL110 Wireless LAN PC Card,

- Xircom CardBus Ethernet 100 + Modem 56 (Ethernet Interface).

Click on Data Sources select 4Dell Latitude --> vol2 -->

WINDOWS\system32\config\software\Microsoft\Windows
NT\CurrentVersion\NetworkCards\

click on descriptions.

Q10 . What is the IP address and MAC address of the computer?

Answer: IP=192.168.1.111, MAC=00:10:a4:93:3e:09

Click on Data Sources select 4Dell Latitude --> vol2 --> Program
[Files/Look@LAN/irunin.ini](#)

Q11 . Search for programs/tools that aided in the crime (Wireless Hacking)

Ans11: The programs which will be used for hacking purpose

1. Look@LAN - An advanced network monitor that allows you to monitor your net in a few clicks.
2. Cain and Abel – A password recovery tool for Microsoft Windows. It can recover many kinds of passwords using methods such as network packet sniffing and cracking various password hashes by using methods such as dictionary attacks, brute force, and cryptanalysis attacks.
3. Network Stumbler - A tool for Windows that facilitates the detection of Wireless LANs using the 802.11b, 802.11a, and 802.11g WLAN standards. It runs on Microsoft Windows operating systems from Windows 2000 to Windows XP.
4. mIRC- An Internet Relay Chat client for Windows, created in 1995.
5. Ethereal/Wireshark - A free and open-source packet analyzer. It is used for network troubleshooting, analysis, software, and communications protocol development, and education.
6. 123WASP - WASP will display all passwords of the currently logged-in user that are stored in the Microsoft PWL file.

Click on Results --> Extracted Content --> Installed Programs

Q12 . Which Email client is used by Mr. Evil?

Answer: Outlook Express, Forte Agent, MSN Explorer, MSN (Hotmail) Email

Click on Data Sources select 4Dell Latitude --> vol2 -->

WINDOWS\system32\config\software\clients\Mail

Q13 . What is the SMTP email address for Mr. Evil?

Answer: iswhoknowsme@sbcglobal.net

Click on Data Sources select 4Dell Latitude --> vol2 --> Program
Files\Agent\Data\Agent.ini

Q14 . How many executable files are in the recycle bin?

Answer: There are 4 files in the recycle bin.

Click on Data Sources select 4Dell Latitude --> vol2 -->Recycler

Q15 . Is there any malware on the computer?

Answer: There is zip bomb malware by the name of unix_hack.giz in this system.

Click on Results --> Extracted Content --> Interesting Items --> Possible Zip Bomb
-->Interesting Files

Q16 . A popular IRC (Internet Relay Chat) program called MIRC was installed.
What is the userid?

Answer: user=Mini Me, email=none@of.ya, nick=Mr, anick=mrevilrulez

How ?

Click on Data Sources select 4Dell Latitude --> vol2 -->Program
Files\mIRC\mirc.ini

Q17 . Ethereal, a popular “sniffing” program that can be used to intercept wired and wireless internet packets was also found to be installed. When TCP packets are collected and re-assembled, the default save directory is the user’s \My Documents directory. What is the name of the file that contains the intercepted data?

Answer: The file name is Interception

How?

Click on Data Sources select 4Dell Latitude --> vol2 --> Document and Settings\Mr.Evil\interception

Q18 . Which internet browser was used?

Answer: Internet Explorer

Click on Data Sources select 4Dell Latitude --> vol2 --> Document and Settings\Mr.Evil\interception

scroll down and see User-Agent: Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC;240x320)

End of the lab

•

•